SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0605 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I **Thamizharasan S** **(192424087)**_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
THAMIZHARASAN S

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process.

(5 Marks)

ANSWER:

Data Collection

|

▼

Data Preprocessing

(Cleaning, Missing Values, Normalization)

|

▼

Feature Extraction

(Select Important Features)

|

▼

Model Training

|

▼

Model Evaluation

(Accuracy, Precision, Recall, etc.)

|

▼

Prediction / Decision

Explanation of Functional Blocks

1. Data Collection
    - Collects data from various sources such as databases, sensors, websites, or files.
    - Example: Collecting customer purchase records for sales prediction.
2. Data Preprocessing
    - Cleans the collected data by removing duplicates, handling missing values, and normalizing data.
    - Improves the quality of data before training.
3. Feature Extraction
    - Selects the most useful attributes (features) from the dataset.
    - Reduces unnecessary information and improves model performance.
    - Example: In email spam detection, features may include the number of links or spam keywords.
4. Model Training
    - The machine learning algorithm learns patterns from the processed data.
    - Examples: Decision Tree, Linear Regression, K-Means.
5. Model Evaluation
    - Measures how well the model performs using metrics such as Accuracy, Precision, Recall, F1-Score, or Mean Squared Error.
    - If performance is satisfactory, the model is used for prediction.

Working of Supervised Learning

- Uses labelled data (input data with known output labels).
- The algorithm learns the relationship between inputs and outputs.

Example:

- Predicting whether an email is Spam or Not Spam.
- Training Data:
    - Email 1 → Spam
    - Email 2 → Not Spam
- After training, the model predicts the label of new emails.

Labelled Data: Available

Working of Unsupervised Learning

- Uses unlabelled data (no predefined output labels).
- The algorithm discovers hidden patterns or groups in the data.

Example:

- Customer segmentation in a shopping mall.
- Customers are automatically grouped based on age, income, and spending habits using K-Means Clustering.

Labelled Data: Not Available

Difference Between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labelled data	Uses unlabelled data
Predicts known outputs	Finds hidden patterns or clusters
Example: Spam Detection	Example:Customer Segmentation

Conclusion

A Machine Learning system collects data, preprocesses it, extracts useful features, trains a model, and evaluates its performance before making predictions. Supervised learning works with labelled data to make predictions, while unsupervised learning works with unlabelled data to discover hidden patterns and group similar data points.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

ANSWER:

Neat Diagram of Disaster Management ML System

Historical Data + Real-Time Sensor Data

|

▼

Data Collection

|

▼

Data Preprocessing

(Cleaning, Missing Values, Normalization)

|

▼

Feature Selection

(Rainfall, River Level, Temperature,

Seismic Activity, Soil Moisture)

|

▼

Model Training

(Neural Network / Random Forest)

|

▼

Model Validation & Evaluation

|

▼

Prediction & Early Warning

|

▼

Emergency Response & Risk Assessment

1. Data Sources

The system collects data from:

- Historical data: Past flood records, earthquake history, rainfall data, river water levels, seismic records.
- Real-time data: Weather stations, IoT sensors, satellite images, river-level sensors, and seismic monitoring stations.

2. Model Selection

Neural Network (Deep Learning) is suitable because:

- Handles very large and complex datasets.
- Learns nonlinear relationships between multiple factors.
- Works well with time-series and sensor data.
- Provides higher prediction accuracy for disaster forecasting.

Alternative Models:

- Random Forest: Good for flood prediction with structured data.
- Decision Tree: Simple and easy to interpret.
- Support Vector Machine (SVM): Suitable for smaller datasets.

Justification:
A Neural Network is preferred because disaster prediction depends on many interconnected factors such as rainfall, river level, humidity, and seismic activity, making it capable of capturing complex patterns.

3. Data Preprocessing

Before training, the data is cleaned and prepared by:

- Removing duplicate records.
- Handling missing values.
- Normalizing numerical values.
- Removing noisy or incorrect data.
- Converting categorical data into numerical format if required.

4. Feature Selection

Important features include:

- Rainfall intensity
- River water level
- Temperature
- Humidity
- Soil moisture
- Seismic wave magnitude
- Ground vibration
- Geographic location

Selecting relevant features reduces computation time and improves prediction accuracy.

5. Model Training, Validation, and Evaluation

Training:

- Train the model using historical disaster data.

Validation:

- Use validation data to tune model parameters and prevent overfitting.

Evaluation:
Measure performance using:

- Accuracy
- Precision
- Recall
- F1-Score

- Confusion Matrix

If the performance is satisfactory, the model is deployed for real-time prediction.

6. Applications in Disaster Management

The machine learning system helps authorities by:

- Providing early warning alerts before floods or earthquakes.
- Identifying high-risk areas for better planning.
- Supporting risk assessment and disaster preparedness.
- Improving emergency response by enabling faster evacuation and resource allocation.
- Reducing loss of life and property through timely predictions.

Conclusion

A machine learning-based disaster management system combines historical records and real-time sensor data to accurately predict disasters. By using a Neural Network (or suitable algorithms like Random Forest), along with proper data preprocessing, feature selection, training, and evaluation, the system enables reliable early warning, risk assessment, and effective emergency response, helping authorities minimize the impact of natural disasters.

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

(5 Marks)

ANSWER:

Neat Diagram of Backpropagation in a Neural Network

Input Layer   Hidden Layer   Output Layer

(x1, x2) -------> (h1, h2) -------> (y)

Forward Pass ▶

Actual Output (Target)

▼

Error Calculation

▼

Backpropagation of Error

▼

Weight Update

▼

Repeat Until Error is Minimum

Backpropagation Algorithm

Backpropagation is a supervised learning algorithm used to train Multilayer Perceptron (MLP) networks. It minimizes the prediction error by adjusting the weights and biases using gradient descent.

Working Process

1. Forward Pass

- Input data is fed into the input layer.
- The inputs pass through hidden layers where weighted sums are calculated.
- An activation function (such as Sigmoid or ReLU) is applied.
- The output layer produces the predicted output.

Example:

- Input: Medical test results.
- Output: Disease = Yes (1) or No (0).

Suppose:

- Predicted output = 0.75
- Actual output = 1

2. Error Calculation

The difference between the predicted and actual output is calculated.

Error = Actual Output − Predicted Output

Example:

- Actual Output = 1
- Predicted Output = 0.75

Error = 1 − 0.75 = 0.25

The objective is to reduce this error as much as possible.

3. Backward Propagation

- The calculated error is propagated from the output layer back to the hidden layer.
- The network computes gradients to determine how much each weight contributed to the error.
- Weights are updated using Gradient Descent.

Weight Update Formula:

New Weight=Old Weight−(Learning Rate×Gradient)

This process is repeated for many iterations (epochs) until the error becomes very small.

Role of Learning Rate

The learning rate (η) controls how much the weights change during each update.

- High Learning Rate
  - Faster training.
  - May overshoot the optimal solution.
  - Can cause unstable training.
- Low Learning Rate
  - More stable learning.
  - Slower convergence.
  - Requires more training time.

A suitable learning rate provides faster and stable convergence.

Role of Activation Functions

Activation functions introduce non-linearity, enabling the neural network to learn complex patterns.

Common activation functions:

- Sigmoid
  - Output range: 0 to 1.
  - Commonly used for binary classification.
- ReLU (Rectified Linear Unit)
  - Faster training.

    - Reduces the vanishing gradient problem.
- Tanh
    - Output range: −1 to 1.
    - Often converges faster than Sigmoid.

The correct activation function improves both accuracy and training efficiency.

Conclusion

Backpropagation trains a Multilayer Perceptron by performing a forward pass, calculating the prediction error, propagating the error backward, and updating the weights using gradient descent. The learning rate determines the speed of learning, while activation functions enable the network to learn complex relationships. Together, they improve the convergence, accuracy, and overall performance of the neural network in binary classification tasks.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution.

(5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

ANSWER:

Initial Dataset (Population – Generation 0)

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Flow of Genetic Algorithm

Initial Population

|

▼

Fitness Evaluation

|

▼

Selection

|

▼

Crossover

|

▼

Mutation

|

▼

New Generation

|

▼

Best Optimal Solution

Step 1: Selection

Selection chooses the individuals with the highest fitness values to become parents for the next generation.

Selected Parents:

- Example 5 → Fitness = 80
- Example 3 → Fitness = 75
- Example 4 → Fitness = 65

Example 1 (Fitness = 40) is rejected because of its low fitness.

Step 2: Crossover

Crossover combines the characteristics of two parent solutions to create new offspring.

Parent 1

- Speed = Medium
- Accuracy = High
- Fitness = 80

Parent 2

- Speed = High
- Accuracy = Medium
- Fitness = 75

Offspring

Offspring	Speed	Accuracy	Estimated Fitness
Child 1	High	High	85
Child 2	Medium	Medium	70

The new child with High Speed and High Accuracy has a better fitness value (85).

Step 3: Mutation

Mutation randomly changes one feature to introduce diversity and avoid getting stuck in a local optimum.

Example:

Before Mutation:

- Speed = High
- Accuracy = High
- Fitness = 85

After Mutation (slight improvement):

- Speed = High

- Accuracy = Very High (or improved High)
- Fitness = 90

Mutation helps discover even better solutions.

Generation-wise Improvement

Generation	Best Fitness
Generation 0	80
Generation 1 (After Selection & Crossover)	85
Generation 2 (After Mutation)	90

The fitness value improves in every generation.

Role of Genetic Algorithm Operations:

Selection

- Chooses the best individuals based on fitness.
- Removes weak solutions.
- Ensures strong candidates survive.

Crossover

- Combines the characteristics of two parents.
- Produces better offspring with improved fitness.
- Explores new combinations of features.

Mutation

- Randomly changes one or more genes.
- Maintains diversity in the population.

- Prevents premature convergence to a local optimum.
- Helps explore the hypothesis space for better solutions.

Final Optimized Solution

Speed	Accuracy	Fitness Value	Suitable
High	High	90	Yes

Conclusion

The Genetic Algorithm starts with an initial population of candidate solutions and repeatedly applies selection, crossover, and mutation to improve their fitness. Selection keeps the best individuals, crossover generates stronger offspring by combining parent characteristics, and mutation introduces diversity to explore new solutions. After successive generations, the algorithm finds the optimal solution with High Speed, High Accuracy, and a maximum fitness value of 90, making it the best suitable solution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

ANSWER:

Dataset

Example	**Age**	**Blood Pressure**	**Glucose Level**	**Lifestyle Risk**	**Actual Output**	**Predicted Output**
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Neural Network for Disease Classification

Input Layer
(Age, Blood Pressure,
Glucose, Lifestyle Risk)
|
▼
Hidden Layer
|
▼
Output Layer
(Disease: Yes / No)
|
▼
Error Calculation

|
▼
Backpropagation
|
▼
Weight Update

Training Using Backpropagation

1. **Input Layer**
   - The patient attributes (Age, Blood Pressure, Glucose Level, Lifestyle Risk) are given as inputs.
2. **Forward Pass**
   - The inputs pass through the hidden layer.
   - The output layer predicts whether the disease is **Yes** or **No**.
3. **Error Calculation**
   - The predicted output is compared with the actual output.
   - The prediction error is calculated.
4. **Backpropagation**
   - The error is propagated backward through the network.
   - The weights are updated using gradient descent to reduce the error.
5. **Repeat**
   - The process is repeated for several epochs until the prediction error becomes minimal.

Confusion Matrix Calculation

Definitions

- **True Positive (TP):** Actual = Yes, Predicted = Yes
- **True Negative (TN):** Actual = No, Predicted = No
- **False Positive (FP):** Actual = No, Predicted = Yes
- **False Negative (FN):** Actual = Yes, Predicted = No

From the Dataset

Example	**Actual**	**Predicted**	**Result**
1	Yes	Yes	**TP**
2	Yes	No	**FN**
3	No	No	**TN**
4	Yes	Yes	**TP**
5	No	Yes	**FP**

Values

- **True Positives (TP) = 2**
- **True Negatives (TN) = 1**
- **False Positives (FP) = 1**
- **False Negatives (FN) = 1**

Evaluation Metrics

1. Accuracy

Accuracy= *TP+TN/TP+TN+FP+FN*

=2+1/2+1+1+1

=3/5

=0.60

Accuracy = 60%

2. Precision

Precision=*TP/TP+FP*

=2/2+1=⅔

=0.667

Precision = 66.7%

3. Recall (Sensitivity)

Recall=$TP/TP+FN$

=2/2+1=⅔

=0.667

Recall = 66.7%

4. Specificity

Specificity=$TN/TN+FP$

=1/1+1=½

=0.50

Specificity = 50%

5. F1-Score

F1=2×Precision×Recall/Precision+Recall

=2×0.667×0.667/0.667+0.667

=0.667

F1-Score = 66.7%

Performance Analysis

Metric	Value
True Positive (TP)	2
True Negative (TN)	1
False Positive (FP)	1
False Negative (FN)	1

Accuracy	**60%**
Precision	**66.7%**
Recall (Sensitivity)	**66.7%**
Specificity	**50%**
F1-Score	**66.7%**

The model correctly predicts **3 out of 5** cases. It performs reasonably well in identifying patients with the disease but has room for improvement, especially in reducing false positives and false negatives.

Role of Preprocessing

Data preprocessing improves model performance by:

- Removing missing or incorrect values.
- Eliminating duplicate records.
- Normalizing numerical data (such as Age).
- Converting categorical values (High, Normal, Low) into numerical values.
- Reducing noise in the dataset.

Role of Feature Selection

Feature selection improves accuracy by choosing the most important attributes, such as:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

Benefits:

- Removes irrelevant features.
- Reduces training time.
- Prevents overfitting.

- Improves prediction accuracy and model reliability.

Conclusion

A neural network classifies disease status by performing a **forward pass**, calculating prediction error, and updating weights through **backpropagation**. From the given dataset, the confusion matrix values are **TP = 2, TN = 1, FP = 1, FN = 1**. The model achieves **60% Accuracy**, **66.7% Precision**, **66.7% Recall**, **50% Specificity**, and **66.7% F1-Score**. Proper **data preprocessing** and **feature selection** further improve the neural network's accuracy, efficiency, and reliability.

6. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

ANSWER:

Traffic Data Collection
(Sensors, CCTV, GPS, Weather)
|
▼
Data Preprocessing
(Cleaning, Missing Values, Normalization)
|
▼
Feature Engineering
(Vehicle Density, Time, Weather,
Road Type, Traffic Signals)
|
▼
Model Training
(Random Forest / Neural Network)
|
▼

Validation & Evaluation

|

▼

Real-Time Prediction

|

▼

Smart Traffic Control & Alerts

1. Data Collection

The system collects data from multiple sources:

- Vehicle density from road sensors and CCTV cameras.
- Time of day (Peak hours/Off-peak hours).
- Weather conditions (Rain, Sunny, Fog).
- Road type (Highway, City Road, Rural Road).
- GPS and traffic signal information.

2. Choice of Algorithm

Recommended Algorithm: Random Forest

Justification:

- Handles both numerical and categorical data.
- Produces high prediction accuracy.
- Resistant to overfitting.
- Works well for traffic congestion classification.

Alternative: Neural Network

- Suitable for very large and complex traffic datasets.
- Learns nonlinear traffic patterns.
- Provides accurate predictions using real-time sensor data.

Conclusion: For most smart traffic systems, Random Forest is preferred because it is accurate, fast, and easy to interpret. Neural Networks are suitable for large-scale intelligent transportation systems.

3. Data Preprocessing

Before training the model:

- Remove duplicate records.
- Handle missing values.
- Normalize numerical features such as vehicle density.
- Convert categorical data (Weather, Road Type) into numerical values.
- Remove noisy and inconsistent data.

These steps improve data quality and model performance.

4. Feature Engineering

Important features include:

- Vehicle density
- Time of day
- Weather conditions
- Road type
- Average vehicle speed
- Traffic signal timings
- Accident reports

Feature engineering removes unnecessary attributes and creates useful features, improving prediction accuracy.

5. Model Training

- Split the dataset into Training (80%) and Testing (20%).
- Train the Random Forest (or Neural Network) model using historical traffic data.
- The model learns patterns that lead to traffic congestion.

6. Validation and Evaluation

The trained model is validated using the test dataset.

Performance is measured using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

If the results are satisfactory, the model is deployed.

7. Real-Time Deployment

The trained model receives live traffic data from sensors and GPS systems.

It predicts:

- Traffic congestion level.
- Travel time.
- Alternative routes.
- Signal timing adjustments.

The predictions are sent to traffic control centers and navigation applications for immediate action.

Applications:

The smart traffic management system helps authorities by:

- Predicting traffic congestion before it becomes severe.
- Optimizing traffic signal timings.
- Suggesting alternate routes to drivers.
- Reducing travel time and fuel consumption.
- Supporting emergency vehicle routing.
- Improving overall road safety.

Conclusion

A machine learning-based smart traffic management system uses vehicle density, time of day, weather conditions, and road type to predict traffic congestion. After data preprocessing and feature engineering, a Random Forest (or Neural Network) model is trained, validated, and deployed using real-time data. The system enables accurate congestion prediction, efficient traffic control, and better decision-making, helping reduce traffic jams and improve transportation efficiency.

7. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

ANSWER

Input Image
|
▼
Convolution Layer
(Feature Extraction)
|
▼
Activation (ReLU)
|
▼
Pooling Layer
(Reduce Image Size)
|
▼
Convolution + Pooling
(Repeated)
|
▼
Fully Connected Layer
|
▼
Output Layer (Class)
(Cat / Dog, Disease / No Disease)

1. Convolutional Layer

- The Convolutional Layer is the first layer of a CNN.

- It applies small filters (kernels) to the input image.
- The filters detect important features such as:
  - Edges
  - Corners
  - Shapes
  - Textures

Output: A set of feature maps.

2. Pooling Layer

- The Pooling Layer reduces the size of the feature maps.
- It decreases computation and memory usage.
- It also helps prevent overfitting.

Types of Pooling:

- Max Pooling: Selects the maximum value from each region.
- Average Pooling: Computes the average value of each region.

3. Fully Connected Layer

- The extracted features are converted into a one-dimensional vector.
- Every neuron is connected to every neuron in the next layer.
- This layer performs the final classification.

Example Output:

- Cat
- Dog
- Car
- Human

4. Feature Map Generation

Feature maps are generated by applying convolution filters to the input image.

Example:

Input Image

Image
|
▼
3×3 Filter
|
▼
Feature Map

Each filter detects different features such as:

- Horizontal edges
- Vertical edges
- Curves
- Textures

As the number of convolution layers increases, the CNN learns more complex features.

5. Backpropagation in CNN

Backpropagation trains the CNN by updating filter weights.

Steps:

1. Perform a forward pass through convolution, pooling, and fully connected layers.
2. Calculate the prediction error by comparing the predicted output with the actual output.
3. Propagate the error backward through the network.
4. Update the filter weights using gradient descent.
5. Repeat until the error becomes minimal.

This process enables the CNN to improve its predictions over multiple training iterations.

6. Example of Image Classification

Suppose a CNN is trained to classify images of Cats and Dogs.

- Input: Image of a cat.
- Convolution Layer: Detects ears, eyes, whiskers, and fur patterns.
- Pooling Layer: Reduces image dimensions while retaining important features.
- Fully Connected Layer: Combines extracted features.
- Output Layer: Predicts Cat with high probability.

If the prediction is incorrect, backpropagation updates the weights to improve future predictions.

7. Factors Affecting Model Performance

The performance of a CNN depends on several factors:

- Quality and quantity of training data.
- Image preprocessing (normalization and resizing).
- Choice of activation function (ReLU is commonly used).
- Learning rate (too high causes instability; too low slows learning).
- Number of convolution and pooling layers.
- Filter (kernel) size.
- Batch size and number of training epochs.
- Regularization techniques such as Dropout to reduce overfitting.

Conclusion

A Convolutional Neural Network (CNN) consists of Convolutional Layers, Pooling Layers, and Fully Connected Layers. Convolutional layers extract important image features, pooling layers reduce dimensionality, and fully connected layers perform classification. Feature maps are generated by applying filters to the input image, while backpropagation updates the filter weights to minimize prediction error. CNNs are widely used in image classification tasks such as face recognition, medical image analysis, and object detection, providing high accuracy when trained with quality data and appropriate model parameters.

8. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity.

(5 Marks)

ANSWER:

Dataset Features
(Age, BP, Glucose, BMI, Cholesterol)
|
▼
Chromosome Representation
(1 = Selected, 0 = Not Selected)
|
▼
Fitness Evaluation
(Classification Accuracy)
|
▼
Selection
|
▼
Crossover
|
▼
Mutation
|
▼
New Feature Subsets
|
▼
Best Optimal Feature Set

1. Chromosome Representation

In a Genetic Algorithm (GA), each candidate feature subset is represented as a binary chromosome, where:

- 1 = Feature is selected.
- 0 = Feature is not selected.

Example:

Features:

- Age
- Blood Pressure
- Glucose
- BMI
- Cholesterol

Chromosome:

1 0 1 1 0

This means the selected features are:

- ✓ Age
- ✓ Glucose
- ✓ BMI

The features Blood Pressure and Cholesterol are excluded.

2. Fitness Evaluation

Each chromosome is evaluated using a machine learning model.

The fitness value is based on:

- Classification accuracy.
- Error rate.
- Number of selected features.

A chromosome with higher accuracy and fewer features receives a higher fitness score.

3. Selection

Selection chooses the chromosomes with the highest fitness values to become parents.

Purpose:

- Keeps the best feature subsets.
- Removes poor-performing solutions.
- Improves the quality of the next generation.

Example:

Chromosome	Fitness
10110	92%
11100	88%
01011	75%

The chromosome with 92% fitness is more likely to be selected.

4. Crossover

Crossover combines two parent chromosomes to create new offspring.

Example:

Parent 1:

10110

Parent 2:

11001

After crossover:

Child 1:

10101

Child 2:

11010

The offspring inherit useful features from both parents, producing better feature combinations.

5. Mutation

Mutation randomly changes one or more bits in a chromosome.

Example:

Before Mutation:

10101

After Mutation:

11101

Mutation:

- Introduces new feature combinations.
- Maintains diversity in the population.
- Prevents the algorithm from getting stuck in a local optimum.

6. Improvement in Model Accuracy and Complexity

Using Genetic Algorithms for feature selection provides several benefits:

- Removes irrelevant and redundant features.
- Improves classification accuracy.
- Reduces model complexity.
- Decreases training time.
- Prevents overfitting.
- Produces a simpler and more efficient machine learning model.

Conclusion

Genetic Algorithms perform feature selection by representing feature subsets as binary chromosomes, evaluating them using a fitness function, and improving them through selection, crossover, and mutation. Over successive generations, the algorithm identifies the optimal feature subset, leading to higher prediction accuracy, reduced computational complexity, and better overall machine learning performance.

9. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.

(5 Marks)

ANSWER:

Sample Confusion Matrix	Predicted: Yes	Predicted: No
Actual: Yes	TP = 50	FN = 10
Actual: No	FP = 5	TN = 35

Evaluation Metrics

1. Accuracy

Accuracy=TP+TN/TP+TN+FP+FN

=50+35/50+35+5+10

=85/100

=0.85

Accuracy = 85%

2. Precision

Precision measures how many predicted positive cases are actually positive.

Precision=(TP/TP+FP)
=50/50+5
=50/55=0.909

Precision = 90.9%

3. Recall (Sensitivity)

Sensitivity=(TP/TP+FN)

=50/50+10
=50/60

=0.833
Recall = 83.3%

4. F1-Score

F1-Score = 2×Precision×Recall/Precision+Recall

=2×0.909×0.833/0.909+0.833
=0.87
F1-Score = 87%

5. ROC Curve (Receiver Operating Characteristic)

- The ROC Curve plots the True Positive Rate (Recall) against the False Positive Rate (FPR) at different classification thresholds.
- It evaluates the model's ability to distinguish between positive and negative classes.
- A model with an Area Under the Curve (AUC) close to 1.0 indicates excellent performance, while an AUC of 0.5 indicates random guessing.

Results

Metric	Formula	Value
Accuracy	(TP + TN) / Total	85%
Precision	TP / (TP + FP)	90.9%
Recall	TP / (TP + FN)	83.3%
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	87%

Importance of Evaluation Metrics

Medical Diagnosis

- Recall is very important because missing a patient with a disease (False Negative) can have serious consequences.
- Precision helps reduce false alarms and unnecessary treatments.
- ROC Curve helps choose the best threshold for diagnosis.

Fraud Detection

- Precision is important to avoid wrongly flagging genuine transactions as fraud.
- Recall ensures that most fraudulent transactions are detected.
- F1-Score provides a balanced measure when both precision and recall are important.
- ROC Curve helps compare different fraud detection models and select the best one.

Conclusion

Evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC Curve are essential for assessing the performance of classification models. Using the given confusion matrix, the model achieved 85% Accuracy, 90.9% Precision, 83.3% Recall, and 87% F1-Score. These metrics help determine how well a model performs in real-world applications like medical diagnosis and fraud detection, where accurate and reliable predictions are critical.

10. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently.

(5 Marks)

ANSWER:

Initial Population
(Different Task Schedules)
|
▼
Chromosome Encoding
|
▼
Fitness Evaluation
(Execution Time, Cost, Resource Use)
|
▼
Selection
|
▼
Crossover
|
▼
Mutation
|
▼
New Generation
|
▼
Best Optimized Schedule

1. Chromosome Encoding

In a Genetic Algorithm (GA), each possible task schedule is represented as a chromosome.

Example:

Suppose four tasks (T1, T2, T3, T4) need to be scheduled.

One chromosome may be:

[T2, T4, T1, T3]

This represents the order in which the tasks are executed.

Different chromosomes represent different scheduling solutions.

2. Fitness Function

The fitness function evaluates how good each schedule is.

It may consider:

- Minimum execution time.
- Maximum CPU utilization.
- Minimum waiting time.
- Minimum resource usage.
- Low execution cost.

A schedule with shorter completion time and better resource utilization receives a higher fitness value.

3. Selection

Selection chooses the best chromosomes based on their fitness values.

Purpose:

- Select high-quality schedules.
- Eliminate poor solutions.
- Pass good characteristics to the next generation.

Example:

Chromosome	Fitness
T2-T4-T1-T3	95
T1-T3-T2-T4	80
T4-T2-T3-T1	70

The chromosome with fitness 95 has the highest chance of being selected.

4. Crossover

Crossover combines two parent chromosomes to create new offspring.

Example:

Parent 1

[T2, T4, T1, T3]

Parent 2

[T1, T3, T2, T4]

Offspring

[T2, T3, T1, T4]

The offspring inherit good scheduling characteristics from both parents, often resulting in improved solutions.

5. Mutation

Mutation randomly changes the position of one or more tasks.

Example:

Before Mutation:

[T2, T3, T1, T4]

After Mutation:

[T2, T1, T3, T4]

Mutation:

- Introduces new scheduling possibilities.
- Maintains diversity in the population.
- Prevents the algorithm from getting stuck in local optima.

6. Finding Near-Optimal Solutions

By repeatedly applying selection, crossover, and mutation, the Genetic Algorithm:

- Improves solution quality in every generation.
- Searches a large solution space efficiently.
- Finds near-optimal task schedules in less time.
- Handles complex optimization problems where exact methods are difficult.

Advantages of Genetic Algorithms

- Efficient for large optimization problems.
- Produces high-quality scheduling solutions.
- Avoids local optimum solutions.
- Reduces execution time and resource utilization.
- Can solve complex real-world scheduling problems.

Conclusion

Genetic Algorithms solve scheduling optimization problems by representing task schedules as chromosomes and evaluating them using a fitness function. The selection process chooses the best schedules, crossover combines good solutions to create better offspring, and mutation introduces diversity to explore new possibilities. Through repeated generations, the algorithm efficiently finds near-optimal scheduling solutions with improved execution time, better resource utilization, and higher overall system performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand